

MODERN

Paper 1: Medicine Through Time with the Western Front

Scholar:

Class:

Teacher:

PROGRESS & ASSESSMENT TRACKER

Lesson / Assessment Title	Effort	Level	Teacher Comments
L1: KT4.1: How have modern discoveries changed our understanding of illness?			
L2: KT4.2: How have prevention and treatment advanced in the modern era?			
L3: KT4.3: How was Penicillin discovered and mass-produced?			
L4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?			
Final Unit Grade:			

L1: KT4.1: How have modern discoveries changed our understanding of illness?

(See Textbook Page 3)

LEARNING OBJECTIVES

- ☐ Explain how the discovery of DNA and the Human Genome Project transformed the understanding of genetic and hereditary illnesses.
- ☐ Analyze the influence of lifestyle choices on modern health and the impact of technology on medical diagnosis.

Do Now Activity

Score: / 10

1. What disease did Edward Jenner vaccine against in 1796?

2. What did Edward Jenner observe that led to his vaccine?

3. Who discovered the cause of cholera in 1854?

4. What structure did Watson and Crick discover in 1953?

5. What is the function of DNA in relation to disease?

6. Name one modern lifestyle factor that causes disease.

7. What did Harvey prove wrong about Galen's theories of blood?

8. Who published the Germ Theory in 1861?

9. Why did the mapping of the human genome (completed in 2003) not lead to immediate cures?

10. Which female scientist's X-ray crystallography photo was vital to Watson and Crick's DNA discovery?

Vocabulary Check

Fill in the blanks using the vocabulary words below.

DNA | Human Genome Project | Biopsy

In the twentieth century, diagnosis shifted from a doctor simply observing a patient's symptoms to rigorous laboratory testing. To identify cancers early, doctors today often perform a _____ on a patient's tissue, examining the sample under powerful microscopes. This transition to evidence-based science was heavily accelerated by breakthroughs in genetics. The 1953 discovery of _____ proved how traits are passed between generations, which eventually led scientists to launch the massive _____ to decode and map the complete set of human genes.

Q1. Identify who discovered the double-helix structure of DNA in 1953.

KNOWLEDGE CHECK (Q2)

How did Watson and Crick's discovery of the double-helix structure of DNA help scientists search for the causes of hereditary diseases?

- ☐ A. It provided a complete blueprint of human biology, allowing scientists to identify specific faulty genes responsible for conditions like cystic fibrosis.
- ☐ B. It proved that hereditary diseases were caused by viruses.
- ☐ C. It allowed scientists to create antibiotics to kill hereditary diseases.
- ☐ D. It proved that hereditary diseases were caused by miasma.

Q3. Describe the goal of the Human Genome Project.

Q4. Explain why Germ Theory could not explain all illnesses.

Q5. Evaluate the impact of the discovery of DNA on modern medicine.

Lesson Consolidation

Reflect on today's learning and answer your teacher's final challenge.

General Notes

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Pair & Share Activity

Q6. Prompt: Which modern development has had a more significant impact on our understanding of the causes of illness: the discovery of genetic factors (DNA and hereditary disease) or the scientific identification of lifestyle factors (smoking, diet, and alcohol)?

Think: Jot down your choice and one piece of evidence from the sources (e.g., how mapping the genome allows women to identify risks for breast cancer, versus how smoking is proven to be the biggest cause of preventable diseases in the world, causing heart disease and lung cancer).

Your Notes:

GCSE Exam Practice

Exam Q34. 'The discovery of DNA was the most significant breakthrough in understanding the causes of illness in the period c1900-present.' How far do you agree? Explain your answer. You may use the following in your answer:

- Watson and Crick
- Lifestyle factors

You must also use information of your own.

[16 marks • 20 mins]

[illegible]

[illegible]

[illegible]

TEACHER FEEDBACK / D.I.R.T.

Exam Q35. Explain why the discovery of the structure of DNA was a significant breakthrough in understanding the causes of illness.

[12 marks • 15 mins]

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Exam Q36. Explain one way in which ideas about the cause of illness in the modern period (c1900-present) were different from ideas in the period c1700-c1900.

[4 marks • 5 mins]

TEACHER FEEDBACK / D.I.R.T.

Exam Q37. 'The discovery of DNA was the most significant breakthrough in understanding the causes of illness in the period c.1900-present.' How far do you agree? Explain your answer.

You may use the following in your answer:

- Watson and Crick
- Lifestyle factors

You must also use information of your own.

L2: KT4.2: How have prevention and treatment advanced in the modern era?

(See Textbook Page 6)

LEARNING OBJECTIVES

- ☐ Explain how scientific and technological developments led to the creation of magic bullets and antibiotics.
- ☐ Analyze the impact of the National Health Service (NHS) on the provision of and access to medical care.

Do Now Activity

Score: / 10

1. What structure did Watson and Crick discover in 1953?

2. What is the function of DNA in relation to disease?

3. Name one modern lifestyle factor that causes disease.

4. What is a 'magic bullet'?

5. Name the first magic bullet, discovered by Paul Ehrlich in 1909.

6. What was established in Britain in 1948 to provide free medical care?

7. What did Edward Jenner vaccinate against in 1796?

8. How did John Snow prove that cholera was spread by contaminated water in 1854?

9. Who was the Minister of Health responsible for establishing the NHS in 1948?

10. Name the second magic bullet, discovered by Gerhard Domagk in 1932 to treat strep infections.

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Magic Bullet Antibiotics National Health Service (NHS)

Your Sentence:

Q1. Identify the first 'Magic Bullet' discovered in 1909.

KNOWLEDGE CHECK (Q2)

What is the key scientific difference between an antiseptic like carbolic acid and a magic bullet like Salvarsan 606?

- ☐ A. Antiseptics are used on the outside of the body to prevent infection, whereas magic bullets are injected into the body to target specific microbes.
- ☐ B. Antiseptics are created from microorganisms, whereas magic bullets are synthetic chemicals.
- ☐ C. Antiseptics are used to cure viruses, whereas magic bullets cure bacteria.
- ☐ D. There is no difference; they are both the same thing.

Q3. Describe how Magic Bullets work.

Q4. Explain how diagnosing illness changed in the 20th century.

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This image shows a single sheet of white paper with horizontal blue lines, resembling notebook paper. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

Q6. Prompt: Which development was the most significant turning point in twentieth-century healthcare: the scientific breakthrough of antibiotics, or the political creation of the National Health Service (NHS)?

Think: Jot down your choice and one piece of evidence from the sources (e.g., how penicillin cured previously fatal infections, or how the NHS removed the fear of poverty by providing free healthcare for millions).

GCSE Exam Practice

Exam Q38. 'Alexander Fleming was the most important individual in the development of penicillin.' How far do you agree? Explain your answer.

[20 marks • 20 mins]

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TEACHER FEEDBACK / D.I.R.T.

Exam Q39. Explain why there was rapid progress in the development of penicillin in the 1940s.

[12 marks • 15 mins]

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Exam Q40. Explain one way in which the treatment of illness in the modern period (c1900-present) was different from the treatment of illness in the period c1700-c1900.

[4 marks • 5 mins]

TEACHER FEEDBACK / D.I.R.T.

Exam Q41. 'Alexander Fleming was the most important individual in the development of penicillin.' How far do you agree? Explain your answer.

[20 marks • 20 mins]

You may use the following in your answer:

- Staphylococcus
- Florey and Chain

You must also use information of your own.

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L3: KT4.3: How was Penicillin discovered and mass-produced?

(See Textbook Page 10)

LEARNING OBJECTIVES

- ☐ Contrast the roles of Alexander Fleming with those of Howard Florey and Ernst Chain in the penicillin breakthrough.
- ☐ Analyze how the factors of government intervention, science and technology, and war collaborated to enable the mass production of penicillin.

Do Now Activity

Score: / 10

1. What is a 'magic bullet'?

2. Name the first magic bullet, discovered by Paul Ehrlich in 1909.

3. What was established in Britain in 1948 to provide free medical care?

4. Who discovered Penicillin by accident in 1928?

5. Which two scientists at Oxford developed Penicillin into a usable drug?

6. What major global event drove the mass production of Penicillin?

7. Who published the Germ Theory in 1861?

8. Who introduced antiseptic surgery using carbolic acid in 1865?

9. What did Fleming observe in his petri dish that led to the discovery of Penicillin?

10. How did the US government assist in the development of Penicillin?

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: **Antibiotic**

Q1. Identify who discovered the first antibiotic in 1928.

Q2. Describe how Penicillin was discovered.

Q3. Explain the roles of Florey and Chain in the development of Penicillin.

Q4. Evaluate the role of World War II in the mass production of Penicillin.

KNOWLEDGE CHECK (Q5)

What was the main scientific and practical limitation that caused Fleming to abandon his work on penicillin in 1931?

- ☐ A. He realized that penicillin was highly toxic to humans.
- ☐ B. He could not secure funding from the government because it was peacetime.
- ☐ C. He lacked the chemical expertise and facilities to extract and purify penicillin in large enough quantities for human trials.
- ☐ D. He was drafted into the army to fight in World War II.

Lesson Consolidation

Reflect on today's learning and answer your teacher's final challenge.

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[illegible]

Q6. Prompt: Who deserves the most credit for the penicillin breakthrough: Alexander Fleming for his accidental discovery in 1928, or Howard Florey and Ernst Chain for their scientific development and therapeutic isolation of the drug in the 1940s?

Think: Jot down your choice and one piece of evidence from the sources (e.g., Fleming recognizing and publishing the mould's bacteria-killing properties in 1928, or the Oxford team actually turning it into a usable drug and proving its clinical effectiveness on humans).

GCSE Exam Practice

Exam Q42. Explain why there was rapid progress in disease prevention in the period c1900-present.

[12 marks • 15 mins]

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TEACHER FEEDBACK / D.I.R.T.

Exam Q43. Explain why the government took a more active role in preventing disease in the period c1900-present.

[12 marks • 15 mins]

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Exam Q44. Explain one way in which ideas about the prevention of illness in the modern period (c1900-present) were different from the medieval period (c1250-c1500).

[4 marks • 5 mins]

TEACHER FEEDBACK / D.I.R.T.

Exam Q45. Explain why there was rapid progress in disease prevention in the period c1900-present.

[12 marks • 15 mins]

You may use the following in your answer:

- Government vaccination campaigns
- Legislation

You must also use information of your own.

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L4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?

(See Textbook Page 13)

LEARNING OBJECTIVES

- ☐ Explain how modern science and technology have transformed the diagnosis and treatment of lung cancer since c1900.
- ☐ Analyze the effectiveness and changing focus of government legislation to prevent smoking-related lung cancer.

Do Now Activity

Score: / 10

1. Who discovered Penicillin by accident in 1928?

2. Which two scientists at Oxford developed Penicillin into a usable drug?

3. What major global event drove the mass production of Penicillin?

4. What lifestyle choice was definitively linked to lung cancer in the 1950s?

5. Name one modern technology used to diagnose lung cancer.

6. State one way the modern government tries to prevent people from smoking.

7. What are the four humours?

8. Who wrote 'On the Fabric of the Human Body' (1543)?

9. Why is lung cancer so difficult to treat effectively compared to other diseases?

10. Which landmark 1950 epidemiological study proved the link between smoking and lung cancer?

Vocabulary Check

Fill in the blanks using the vocabulary words below.

Bronchoscopy | Immunotherapy | Nanny State

In twenty-first-century Britain, the diagnosis and treatment of lung cancer have been transformed by science and technology. Rather than relying on simple chest X-rays, doctors can perform a highly precise _____ to examine the inside of the lungs and collect cell samples for laboratory testing. When tumours are detected, patients are often treated with cutting-edge therapies such as _____ to trigger their body's natural defenses to destroy cancer cells. At the same time, the government has passed highly restrictive laws to prevent people from smoking, leading some critics to complain that state interference in lifestyle choices represents the rise of an overbearing _____ .

Q1. Identify one method used to diagnose Lung Cancer.

Q2. Describe the aggressive treatments used for Lung Cancer.

Q3. Explain why the UK government has focused heavily on preventing Lung Cancer.

Q4. Evaluate the effectiveness of government legislation in reducing smoking.

KNOWLEDGE CHECK (Q5)

What is the key clinical advantage of using a CT scan over a traditional chest X-ray when diagnosing a lung cancer patient?

- ☐ A. A CT scan uses harmless sound waves instead of radiation.
- ☐ B. A CT scan can be performed at home by the patient.
- ☐ C. A CT scan creates highly detailed, 3D cross-sectional images of the lungs, allowing doctors to detect much smaller tumors than a 2D X-ray.
- ☐ D. A CT scan can actively cure the lung cancer by destroying the tumor.

Lesson Consolidation

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Q6. Prompt: Which category of modern government action has been the most significant turning point in the prevention of lung cancer: directly warning and educating current smokers (such as plain packaging and graphic warning labels), preventing young people from starting (such as raising the legal age of sale from 16 to 18), or protecting non-smokers from second-hand smoke (such as the 2007 indoor smoking ban)?

Think: Jot down your choice and one piece of evidence from your knowledge (for instance, the 2007 ban significantly shifting social attitudes about where smoking is acceptable, or how 1950s research first proved the definitive link between tobacco and cancer).

GCSE Exam Practice

Exam Q46. Explain why the government took action to improve public health in the period c1900-present.

[12 marks • 15 mins]

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TEACHER FEEDBACK / D.I.R.T.

Exam Q47. Explain why the creation of the National Health Service (NHS) in 1948 improved public health in Britain.

[12 marks • 15 mins]

[illegible]

Exam Q48. Explain one way in which hospital care in the modern period (c1900-present) was different from hospital care in the period c1700-c1900.

[4 marks • 5 mins]

TEACHER FEEDBACK / D.I.R.T.

Exam Q49. Explain why the government took action to improve public health in the period c1900-present.

[12 marks • 15 mins]

You may use the following in your answer:

- Lung cancer
- The NHS

You must also use information of your own.

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Factors Overview: Modern

Edexcel focuses heavily on the factors that drove medical progress (or held it back). For each factor below, write one specific historical example from this period that either helped or hindered medical progress.

Factor	Specific Historical Example & Impact
Individuals	
The Church & Religion	
Government & Wealth	
Science & Technology	
Attitudes in Society	
War	

